CHAPTER

2

The global energy balance

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We consider now the general problem of the radiative equilibrium temperature of the Earth. The Earth is bathed in solar radiation and absorbs much of that incident upon it. To maintain equilibrium it must warm up and radiate energy away at the same rate it is received, as depicted in Fig. 2.1. We will see that the emission temperature of the Earth is 255 K and that a body at this temperature radiates energy primarily in the infrared (IR). But the atmosphere is strongly absorbing at these wavelengths due to the presence of trace gases—principally the triatomic molecules H_2O and CO_2—which absorb and emit in the infrared, thus raising the surface temperature above that of the emission temperature, a mechanism that has become known as the *greenhouse effect*.

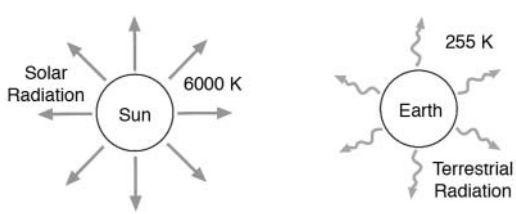

FIGURE 2.1. The Earth radiates energy at the same rate it is received from the Sun. The Earth's emission temperature is 255 K, and that of the Sun is 6000 K. The outgoing terrestrial radiation peaks in the infrared spectrum; the incoming solar radiation peaks at shorter wavelengths, in the visible spectrum.

2.1. PLANETARY EMISSION TEMPERATURE

The Earth receives almost all of its energy from the Sun. At the present time in its evolution the Sun emits energy at a rate of $Q = 3.87 \times 10^{26}$ W. The flux of solar energy at the Earth, called the *solar constant*, depends on the distance of the Earth from the Sun, r, and is given by the inverse square law, $S_0 = Q/4\pi r^2$. Of course, because of variations in the Earth's orbit (see Sections 5.1.1 and 12.3.5) the solar constant is not really constant; the terrestrial value

TABLE 2.1. Properties of some of the planets. S_0 is the solar constant at a distance r from the Sun, α_p is the planetary albedo, T_e is the emission temperature computed from Eq. 2-4, T_m is the measured emission temperature, and T_s is the global mean surface temperature. The rotation period, τ, is given in Earth days.

	r 10^9 m	S_0 W m^{-2}	α_p	T_e K	T_m K	T_s K	τ Earth days
Venus	108	2632	0.77	227	230	760	243
Earth	150	1367	0.30	255	250	288	1.00
Mars	228	589	0.24	211	220	230	1.03
Jupiter	780	51	0.51	103	130	134	0.41

$S_0 = 1367\,\mathrm{Wm^{-2}}$, set out in Table 2.1, along with that for other planets, is an average corresponding to the average distance of Earth from the Sun, $r = 150 \times 10^9$ m.

The way radiation interacts with an atmosphere depends on the wavelength as well as the intensity of the radiative flux. The relation between the energy flux and wavelength, which is the spectrum, is plotted in Fig. 2.2. The Sun emits radiation that is primarily in the visible part of the spectrum, corresponding to the colors of the rainbow—red, orange, yellow, green, blue, indigo and violet—with the energy flux decreasing toward longer (infrared, IR) and shorter (ultraviolet, UV) wavelengths.

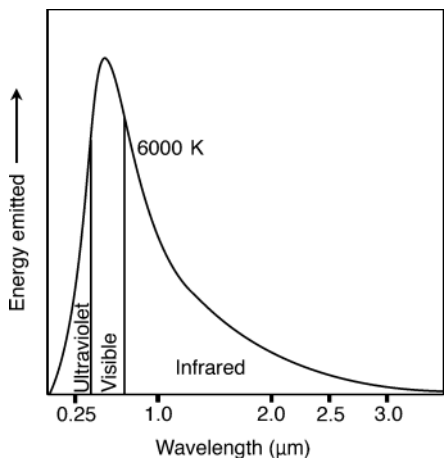

FIGURE 2.2. The energy emitted from the Sun plotted against wavelength based on a black body curve with $T = T_{Sun}$. Most of the energy is in the visible spectrum, and 95% of the total energy lies between 0.25 and 2.5μm (10^{-6}m).

Why does the spectrum have this pattern? Such behavior is characteristic of the radiation emitted by incandescent material, as can be observed for example in a coal fire. The hottest parts of the fire are almost white and emit the most intense radiation, with a wavelength that is shorter than that coming from the warm parts of the fire, which glow red. The coldest parts of the fire do not seem to be radiating at all, but are, in fact, radiating in the infrared. Experiment and theory show that the wavelength at which the intensity of radiation is maximum, and the flux of emitted radiation, depend only on the temperature of the source. The theoretical spectrum, one of the jewels of physics, was worked out by Planck,[1] and is known as the *Planck* or

[1]

In 1900 Max Planck (1858–1947) combined the formulae of Wien and Rayleigh, describing the distribution of energy as a function of wavelength of the radiation in a cavity at temperature T, to arrive at what is now known as Planck's radiation curve. He went on to a complete theoretical deduction, introduced quanta of energy, and set the scene for the development of quantum mechanics.

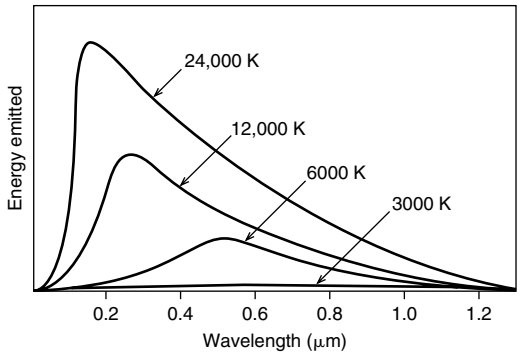

FIGURE 2.3. The energy emitted at different wavelengths for blackbodies at several temperatures. The function $B_\lambda(T)$, Eq. A-1, is plotted.

blackbody spectrum. (A brief theoretical background to the Planck spectrum is given in Appendix A.1.1.) It is plotted as a function of temperature in Fig. 2.3. Note that the hotter the radiating body, the more energy it emits at shorter wavelengths. If the observed radiation spectrum of the Sun is fitted to the blackbody curve by using T as a free parameter, we deduce that the blackbody temperature of the Sun is about 6000 K.

Let us consider the energy balance of the Earth as in Fig. 2.4, which shows the Earth intercepting the solar energy flux and radiating terrestrial energy. If at the location of the (mean) Earth orbit, the incoming solar energy flux is $S_0 = 1367\ \mathrm{W\,m^{-2}}$, then, given that the cross-sectional area of the Earth intercepting the solar energy flux is πa^2, where a is the radius of the Earth (Fig. 2.4),

$$\text{solar power incident on the Earth} = S_0 \pi a^2 = 1.74 \times 10^{17}\ \mathrm{W},$$

using the data in Table 1.1. Not all of this radiation is absorbed by the Earth; a significant fraction is reflected. The ratio of reflected to incident solar energy is called the *albedo*, α. As set out in Table 2.2 and the map of surface albedo shown in Fig. 2.5, α depends on the nature of the reflecting surface and is large for clouds, light surfaces such as deserts, and (especially) snow and ice. Under the present terrestrial conditions of cloudiness and snow and ice cover, on average a fraction $\alpha_p \simeq 0.30$ of the incoming solar radiation at the Earth is reflected back to space; α_p is known as the *planetary albedo* (see Table 2.1). Thus

TABLE 2.2. Albedos for different surfaces. Note that the albedo of clouds is highly variable and depends on the type and form. See also the horizontal map of albedo shown in Fig. 2.5.

Type of surface	Albedo (%)
Ocean	2–10
Forest	6–18
Cities	14–18
Grass	7–25
Soil	10–20
Grassland	16–20
Desert (sand)	35–45
Ice	20–70
Cloud (thin, thick stratus)	30, 60–70
Snow (old)	40–60
Snow (fresh)	75–95

$$\text{Solar radiation absorbed by the Earth} = (1 - \alpha_p) S_0 \pi a^2 = 1.22 \times 10^{17}\ \mathrm{W}. \quad (2\text{-}1)$$

In equilibrium, the total terrestrial flux radiated to space must balance the solar radiation absorbed by the Earth. If in total the spinning Earth radiates in all directions like a blackbody of uniform temperature T_e (known as the *effective planetary temperature*, or *emission temperature* of the Earth) the Stefan-Boltzmann law gives:

$$\text{Emitted radiation per unit area} = \sigma T_e^4 \quad (2\text{-}2)$$

where $\sigma = 5.67 \times 10^{-8}\mathrm{W\,m^{-2}\,K^{-4}}$ is the Stefan-Boltzmann constant. So

$$\text{Emitted terrestrial radiation} = 4\pi a^2 \sigma T_e^4. \quad (2\text{-}3)$$

Note that Eq. 2-3 is a **definition** of *emission temperature* T_e. It is the temperature one would infer by looking back at Earth if a blackbody curve was fitted to the measured spectrum of outgoing radiation.

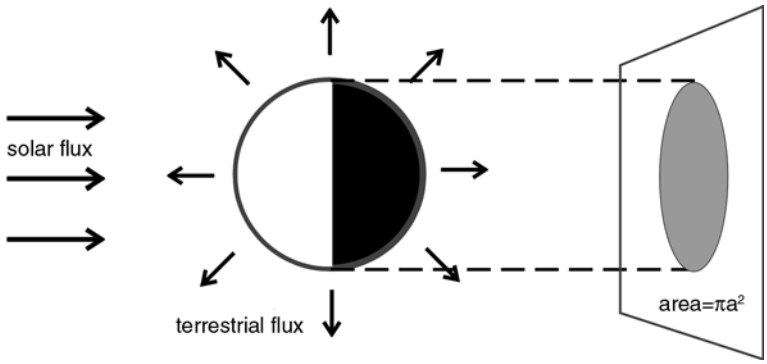

FIGURE 2.4. The spinning Earth is imagined to intercept solar energy over a disk of radius a and radiate terrestrial energy away isotropically from the sphere. Modified from Hartmann, 1994.

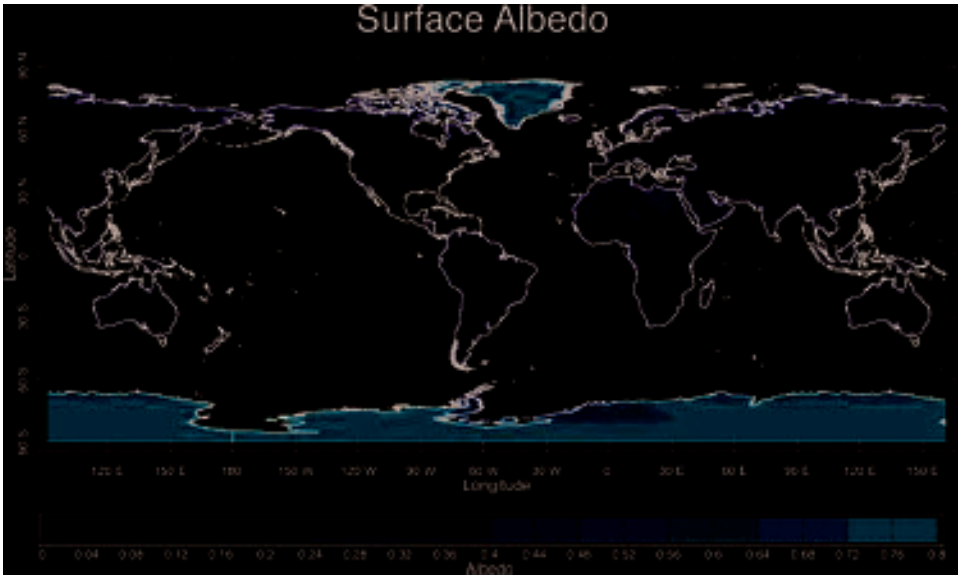

FIGURE 2.5. The albedo of the Earth's surface. Over the ocean the albedo is small (2–10%). It is larger over the land (typically 35–45% over desert regions) and is particularly high over snow and ice (~80%) (see Table 2.2).

Equating Eq. 2-1 with Eq. 2-3 gives

$$T_e = \left[\frac{S_0(1-\alpha_p)}{4\sigma}\right]^{1/4}. \qquad (2\text{-}4)$$

Note that the radius of the Earth has cancelled out: T_e depends only on the planetary albedo and the distance of the Earth from the Sun. Putting in numbers, we find that the Earth has an emission temperature of 255 K. Table 2.1 lists the various parameters for some of the planets and compares approximate measured values, T_m, with T_e computed from Eq. 2-4. The agreement is very good, except for Jupiter where it is thought that about one half of the energy input comes from the gravitational collapse of the planet (see Problem 3 at end of this chapter).

However, as can be seen from Table 2.1, the emission temperature of Earth is nearly 40 K cooler than the globally averaged observed surface temperature, which is $T_s = 288$ K. As we shall discuss in Section 2.3, $T_s \neq T_e$ because: (1) radiation is absorbed within the atmosphere,

principally by its water vapor blanket, and (2) fluid motions—air currents—carry heat both vertically and horizontally.

2.2. THE ATMOSPHERIC ABSORPTION SPECTRUM

A property of the blackbody radiation curve is that the wavelength of maximum energy emission, λ_m, satisfies

$$\lambda_m T = \text{constant}. \tag{2-5}$$

This is known as *Wien's displacement law*. Since the solar emission temperature is about 6000 K, the maximum of the solar spectrum is (see Fig. 2.2) at about 0.6 μm (in the visible spectrum), and we have determined $T_e = 255$ K for the Earth, it follows that the peak of the terrestrial spectrum is at

$$\lambda_m^{Earth} = 0.6\,\mu\text{m} \times \frac{6000}{255} \simeq 14\,\mu\text{m}.$$

Thus the Earth's radiation to space is primarily in the infrared spectrum. Normalized (see Appendix A.1.1) blackbody spectra for the Sun and Earth are shown in Fig. 2.6. The two spectra hardly overlap, which greatly simplifies thinking about radiative transfer.

Also shown in Fig. 2.6 is the atmospheric absorption spectrum; this is the fraction of radiation at each wavelength that is

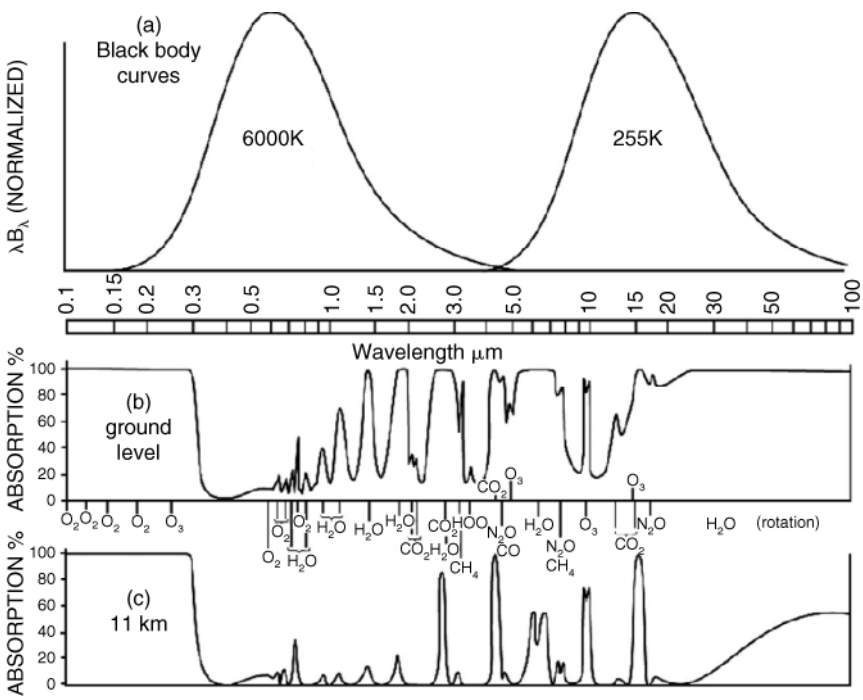

FIGURE 2.6. (a) The normalized blackbody emission spectra, $T^{-4}\lambda B_\lambda$, for the Sun ($T = 6000$ K) and Earth ($T = 255$ K) as a function of ln λ (top), where B_λ is the blackbody function (see Eq. A-2) and λ is the wavelength (see Appendix A.1.1 for further discussion). (b) The fraction of radiation absorbed while passing from the ground to the top of the atmosphere as a function of wavelength. (c) The fraction of radiation absorbed from the tropopause (typically at a height of 11 km) to the top of the atmosphere as a function of wavelength. The atmospheric molecules contributing the important absorption features at each frequency are also indicated. After Goody and Yung (1989).

absorbed on a single vertical path through the atmosphere. From it we see that:

- The atmosphere is almost completely transparent in the visible spectrum, at the peak of the solar spectrum.
- The atmosphere is very opaque in the UV spectrum.
- The atmosphere has variable opacity across the IR spectrum. It is almost completely opaque at some wavelengths, and transparent at others.
- N_2 does not figure at all in absorption, and O_2 absorbs only in the far UV (where there is little solar energy flux) and, a little, in the near IR. The dominant constituents of the atmosphere are incredibly transparent across almost the whole spectral range of importance.
- The absorption of terrestrial radiation is dominated by triatomic molecules—O_3 in the UV; H_2O, CO_2, and others in the IR—because triatomic molecules have rotational and vibrational modes that can easily be excited by IR radiation. These molecules are present in tiny concentrations (see Table 1.2) but play a key role in the absorption of terrestrial radiation (see Fig. 2.6). They are known as *greenhouse gases*. This is the fundamental reason why atmospheric radiation may be so vulnerable to the human-induced changes in composition shown in Fig. 1.3.

2.3. THE GREENHOUSE EFFECT

The global average mean surface temperature of the Earth is 288 K (Table 2.1). Previously we deduced that the emission temperature of the Earth is 255 K, which is considerably lower. Why? We saw from Fig. 2.6 that the atmosphere is rather opaque to IR radiation, so we cannot think of terrestrial radiation as being radiated into space directly from the surface. Much of the radiation emanating from the surface will be absorbed, primarily by H_2O, before passing through the atmosphere. On average, the emission to space will emanate from some level in the atmosphere (typically about 5 km) such that the region above that level is mostly transparent to IR radiation. It is this region of the atmosphere, rather than the surface, that must be at the emission temperature. Thus radiation from the atmosphere will be directed downward as well as upward, and hence the surface will receive not only the net solar radiation, but IR from the atmosphere as well. Because the surface feels more incoming radiation than if the atmosphere were not present (or were completely transparent to IR) it becomes warmer than T_e. This has become known as the *greenhouse effect*.[2]

2.3.1. A simple greenhouse model

Consider Fig. 2.7. Since the atmosphere is thin, let us simplify things by considering a planar geometry, in which the incoming radiation per unit area is equal to the average flux per unit area striking the Earth. This average incoming solar energy *per unit area of the Earth's surface* is

$$\text{average solar energy flux} = \frac{\text{intercepted incoming radiation}}{\text{Earth's surface area}} = \frac{S_0 \pi a^2}{4\pi a^2} = \frac{S_0}{4}. \tag{2-6}$$

We will represent the atmosphere by a single layer of temperature T_a, and, in this first calculation, assume: (1) that it is completely transparent to shortwave solar radiation, and (2) that it is completely opaque to IR radiation (i.e., it absorbs all the IR radiating

[2]It is interesting to note that the domestic greenhouse does not work in this manner! A greenhouse made of plastic window panes, rather than conventional glass, is effective even though plastic (unlike glass) does not have significant absorption bands in the IR. The greenhouse works because its windows allow energy in and its walls prevent the warm air from rising or blowing away.